

S4 | Inversa unei matrice. Sisteme de ec. liniare

ex: c) $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$; $\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & -1 & -1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 & 1 & 0 \\ 1 & -1 & -1 & 1 & 0 & 0 & 0 & 1 \end{array} \right]$

$$\Rightarrow A^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = \frac{1}{4} A$$

d) $A = \begin{bmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 2 \end{bmatrix}$, $\alpha \in \mathbb{R}$ $\left[\begin{array}{ccc|ccc} 2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & \alpha & 0 & 0 & 1 \end{array} \right] \xrightarrow{\frac{1}{2}l_1}$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & \alpha & 0 & 0 & 1 \end{array} \right] \xrightarrow{\begin{matrix} l_2 - l_1 \\ l_3 + l_1 \end{matrix}} \left[\begin{array}{ccc|ccc} 1 & 1 & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 0 & -2 & -\frac{3}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & 3 & \alpha + \frac{3}{2} & \frac{1}{2} & 0 & 1 \end{array} \right] \xrightarrow{-\frac{1}{2}l_2}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & \frac{3}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 1 & \frac{3}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 3 & \alpha + \frac{3}{2} & \frac{1}{2} & 0 & 1 \end{array} \right] \xrightarrow{\begin{matrix} l_1 - l_2 \\ l_3 - 2l_2 \end{matrix}} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{4} & \frac{1}{4} & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{3}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 0 & \alpha - \frac{3}{4} & -\frac{1}{4} & \frac{3}{2} & 1 \end{array} \right]$$

Dacă $\alpha = \frac{3}{4} \Rightarrow A$ nu e inversabilă

Dacă $\alpha \neq \frac{3}{4} \Rightarrow$

$$\xrightarrow{l_3 - \frac{1}{4\alpha-3}l_3} \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{3}{4} & \frac{1}{4} & -\frac{1}{2} & 0 \\ 0 & 1 & \frac{3}{4} & \frac{1}{4} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -\frac{1}{4\alpha-3} & \frac{6}{4\alpha-3} & \frac{4}{4\alpha-3} \end{array} \right], \text{ not } \frac{1}{4\alpha-3} := \beta$$

$$\xrightarrow{\begin{matrix} l_1 - \frac{3}{4}l_3 \\ l_2 - \frac{3}{4}l_3 \end{matrix}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \alpha\beta & (2\alpha-6)\beta & -3\beta \\ 0 & 1 & 0 & \alpha\beta & -(2\alpha+3)\beta & -3\beta \\ 0 & 0 & 1 & -\beta & 6\beta & 4\beta \end{array} \right] \xrightarrow{\begin{matrix} y_3 \\ A^{-1} \end{matrix}} \Rightarrow A^{-1} = \frac{1}{4\alpha-3} \begin{bmatrix} \alpha & 2\alpha-6 & -3 \\ \alpha & -(2\alpha+3) & -3 \\ -1 & 6 & 4 \end{bmatrix}$$

2

→ de ex tunc a maximum ordinis

sist este
compatibil
nedeterminat

2

Solutive =

$$\left\{ \frac{1}{3}(1-x_3-4x_2), x_2, x_3 \in \mathbb{R} \right\}$$

c)
$$\begin{cases} x_1 + x_2 - 3x_3 = -1 \\ 2x_1 + x_2 - 2x_3 = 1 \\ x_1 + x_2 + x_3 = 3 \\ x_1 + 2x_2 - 3x_3 = 1 \end{cases} \quad \text{In } \mathbb{R}^3, \bar{A} = \begin{bmatrix} 1 & 1 & -3 & -1 \\ 2 & 1 & -2 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & -3 & 1 \end{bmatrix} \begin{matrix} l_2 - 2l_1 \\ l_3 - l_1 \\ l_4 - l_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & -3 & -1 \\ 0 & -1 & 4 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix} \xrightarrow{l_4 + l_2} \begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & -1 & 4 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 4 & 5 \end{bmatrix} \xrightarrow{l_4 - l_3} \begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & -1 & 4 & 3 \\ 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ sist este incompatibil

d)
$$\begin{cases} 5x_1 - 3x_2 + 2x_3 + 4x_4 = 3 \\ 4x_1 - 2x_2 + 3x_3 + 7x_4 = 1 \\ 8x_1 - 6x_2 - x_3 - 5x_4 = 9 \\ 7x_1 - 3x_2 + 7x_3 + 17x_4 = \alpha \end{cases} \quad \text{In } \mathbb{R}^4 \quad \alpha \in \mathbb{R}$$

$$\bar{A} = \begin{bmatrix} 5 & -3 & 2 & 4 & 3 \\ 4 & -2 & 3 & 7 & 1 \\ 8 & -6 & -1 & -5 & 9 \\ 3 & -3 & 7 & 17 & \alpha \end{bmatrix} \xrightarrow{l_1 - l_2} \begin{bmatrix} 1 & -1 & -1 & -3 & 2 \\ 4 & -2 & 3 & 7 & 1 \\ 8 & -6 & -1 & -5 & 9 \\ 3 & -3 & 7 & 17 & \alpha \end{bmatrix} \begin{matrix} l_2 - 4l_1 \\ l_3 - 8l_1 \\ l_4 - 3l_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -1 & -1 & -3 & 2 \\ 0 & 2 & 7 & 19 & -7 \\ 0 & 2 & 7 & 19 & -7 \\ 0 & 4 & 14 & 38 & \alpha - 14 \end{bmatrix} \begin{matrix} l_3 - l_2 \\ l_4 - 2l_2 \end{matrix} \xrightarrow{\text{see same}} \begin{bmatrix} 1 & -1 & -1 & -3 & 2 \\ 0 & 2 & 7 & 19 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \alpha \end{bmatrix}$$

Dacă: $\alpha \neq 0$ sistemul este incompatibil

Dacă: $\alpha = 0$ sistemul este comp medet $\Leftrightarrow$

$$\Leftrightarrow \begin{cases} x_1 - x_2 - x_3 - 3x_4 = 2 \\ 2x_2 + 7x_3 + 19x_4 = -7 \end{cases} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} x_1 - x_2 = 2 + x_3 + 3x_4 \\ 2x_2 = -7 - 7x_3 - 19x_4 \Rightarrow x_2 = -\frac{7}{2} - \frac{7}{2}x_3 - \frac{19}{2}x_4 \end{cases}$$

$$x_1 = 2 + x_3 + 3x_4 - \frac{7}{2} - \frac{7}{2}x_3 - \frac{19}{2}x_4 = -\frac{3}{2} - \frac{5}{2}x_3 - \frac{13}{2}x_4$$

$$\Rightarrow \left(\text{Solutiile: } \left(-\frac{3}{2} - \frac{5}{2}x_3 - \frac{13}{2}x_4, -\frac{7}{2} - \frac{7}{2}x_3 - \frac{19}{2}x_4, x_3, x_4 \right) \right)$$

e) $\begin{cases} \alpha x_1 + x_2 + x_3 = 1 \\ x_1 + \alpha x_2 + x_3 = 1 \\ x_1 + x_2 + \alpha x_3 = 1 \end{cases} \quad \text{în } \mathbb{R}^3$

$$\bar{A} = \left[\begin{array}{ccc|c} \alpha & 1 & 1 & 1 \\ 1 & \alpha & 1 & 1 \\ 1 & 1 & \alpha & 1 \end{array} \right] \xrightarrow{l_3 \leftrightarrow l_1} \left[\begin{array}{ccc|c} 1 & 1 & \alpha & 1 \\ 1 & \alpha & 1 & 1 \\ \alpha & 1 & 1 & 1 \end{array} \right] \xrightarrow{\substack{l_2 - l_1 \\ l_3 - \alpha l_1}}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & \alpha & 1 \\ 0 & \alpha-1 & 1-\alpha & 0 \\ 0 & 1-\alpha & 1-\alpha^2 & 1-\alpha \end{array} \right] \xrightarrow{l_3 + l_2} \left[\begin{array}{ccc|c} 1 & 1 & \alpha & 1 \\ 0 & \alpha-1 & 1-\alpha & 0 \\ 0 & 0 & 2-\alpha^2-\alpha & 1-\alpha \end{array} \right]$$

$(1-\alpha)(2+\alpha)$

Dacă $\alpha = 1$ $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$ sist comp medet.

$$(x_1 + x_2 + x_3) = 1$$

$$x_1 = 1 - x_2 - x_3$$

Soluția $\{ (1-x_2-x_3, x_2, x_3) \} \mid x_2, x_3 \in \mathbb{R}$

Deci $\alpha = 2 \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 3 \end{array} \right] \Rightarrow \text{sist este incompatibil}$

Deci $\alpha \in \mathbb{R} \setminus \{1, -2\}$: sist complet det.

$$\Leftrightarrow \begin{cases} x_1 + x_2 + \alpha x_3 = 1 \\ (\alpha - 1)x_2 + (1 - 2)x_3 = 0 \\ (1 - \alpha)(2 + \alpha)x_3 = 1 - \alpha \end{cases}$$

$$\Leftrightarrow \begin{cases} x_1 + x_2 + \alpha x_3 = 1 \Rightarrow x_1 = 1 - \frac{1}{2 + \alpha} - \frac{\alpha}{2 + \alpha} \\ -x_2 + x_3 = 0 \Rightarrow x_2 = \frac{1}{2 + \alpha} \\ (2 + \alpha)x_3 = 1 \Rightarrow x_3 = \frac{1}{2 + \alpha} \end{cases} \quad \downarrow \\ = \frac{1}{2 + \alpha}$$

$\Rightarrow$ Soluțiile: $\left\{ \left(\frac{1}{2 + \alpha}, \frac{1}{2 + \alpha}, \frac{1}{2 + \alpha} \right) \right\}$